Illustration 22: Two numbers when divided by a certain divisor give remainders 473 and 298, respectively. When their sum is divided by the same divisor, the remainder is 236. Find the divisor.

Solution: The required divisor

$= 437 + 298 - 236 = 499.$

EXERCISE-I

1. $7372 \times 7372 + 7372 \times 628 = ?$

(a) 58976000 (b) 58967000
(c) 5897600 (d) None of these

2. $9999 + 8888 + 777 + ? = 19700$

(a) 36 (b) 16
(c) 64 (d) 26

3. $60\ ?\ 6 \times 111 = 666666$

(a) 0 (b) 2
(c) 1 (d) 6

4. $3149 \times 1\ ?\ 5 = 425115$

(a) 3 (b) 2
(c) 4 (d) 6

5. If the two digits of the age of Mr Manoj are reversed then the new age so obtained is the age of his wife. $\frac{1}{11}$ of the sum of their ages is equal to the difference between their ages. If Mr Manoj is older than his wife then find the difference between their ages.

(a) Cannot be determined
(b) 8 years
(c) 10 years
(d) 9 years
(e) 7 years

6. If in a long division sum, the dividend is 380606 and the successive remainders from the first to the last are 434, 125 and 413, then divisor is:

(a) 451 (b) 843
(c) 4215 (d) 3372

7. If $\frac{x}{y} = \frac{3}{4}$, then the value of $\left(\frac{6}{7} + \frac{y-x}{y+x}\right)$ equals:

(a) $\frac{5}{7}$ (b) $1\frac{1}{7}$
(c) 1 (d) 2

8. The largest natural number by which the product of three consecutive even natural numbers is always divisible, is:

(a) 16 (b) 24
(c) 48 (d) 96

9. Which number should replace both the '*'s in $\left(\frac{*}{21}\right) \times \left(\frac{*}{189}\right) = 1$?

(a) 21 (b) 63
(c) 3969 (c) 147

10. In a division sum, the divisor is 12 times the quotient and 5 times the remainder. If the remainder be 48, then the dividend is:

(a) 240 (b) 576
(c) 4800 (d) 4848

11. What least number must be subtracted from 1294 so that the remainder when divided by 9, 11, 13 will leave in each case the same remainder 6?

(a) 0 (b) 1
(c) 2 (d) 3

12. 24 is divided into two parts such that 7 times the first part added to 5 times the second part makes 146. The first part is:

(a) 11 (b) 13
(c) 16 (d) 17

13. $\frac{1}{4}$ of a number subtracted from $\frac{1}{3}$ of the same number gives 12. The number is:

(a) 144 (b) 120
(c) 72 (d) 63

14. $\frac{4}{3}$ of a certain number is 64. Half of that number is:

(a) 32 (b) 40
(c) 80 (d) 16

15. A fraction becomes 4 when 1 is added to both the numerator and denominator; and it becomes 7 when 1 is subtracted from both the numerator and denominator. The numerator of the given fraction is:

(a) 2 (b) 3
(c) 7 (d) 15

16. Three numbers are in the ratio 3:4:5. The sum of the largest and the smallest equals the sum of the third and is 52. The smallest number is:

(a) 20 (b) 27
(c) 39 (d) 52

17. The sum of three numbers is 68. If the ratio between first and second is 2:3 and that between second and third is 5:3, then the second number is:

(a) 30 (b) 20
(c) 58 (d) 48

18. If 1 is added to the denominator of a fraction, the fraction becomes $\frac{1}{2}$. If 1 is added to the numerator, the fraction becomes 1. The fraction is:

(a) $\frac{4}{7}$ (b) $\frac{5}{9}$

(c) $\frac{2}{3}$ (c) $\frac{10}{11}$

19. $\frac{4}{5}$ of a number exceeds its $\frac{2}{3}$ by 8. The number is:

(a) 30 (b) 60
(c) 90 (d) None of these

20. What is the sum of all prime numbers from 60 to 80?

(a) 361 (b) 341
(c) 351 (d) 349

21. The quotient arising from the division of 24446 by a certain divisor is 79 and the remainder is 35, what is the divisor?

(a) 39 (b) 309
(c) 390 (d) 3009

22. In a division sum, the quotient is 120, the divisor 456 and the remainder 333, find the dividend.

(a) 5533 (b) 50553
(c) 56053 (d) 55053

23. The quotient arising from a division of a number by 62 is 463 and the remainder is 60, what is the number?

(a) 28666 (b) 28766
(c) 28576 (d) 28676

24. A number when divided by 221 gives a remainder 43. What remainder will be obtained by dividing the same number by 17?

(a) 11 (b) 8
(c) 9 (d) 13

25. Which one of the following is the largest prime number of three digits?

(a) 997 (b) 999
(c) 991 (d) 993

26. When a certain number is multiplied by 7, the product consists entirely of fives; find the least value of such a number.

(a) 79365 (b) 78365
(c) 77365 (d) 79265

27. In a division sum, the divisor is 10 times the quotient and five times the remainder. What is the dividend, if the remainder is 46?

(a) 5636 (b) 5536
(c) 5336 (d) 5436

28. Which one of the following is the least number of four digits divisible by 71?

(a) 1006 (b) 1065
(c) 1094 (d) 1056

29. How many numbers up to 100 are divisible by 7?

(a) 14 (b) 107
(c) 93 (d) 100

30. How many numbers up to 500 are divisible by 23?

(a) 23 (b) 27
(c) 21 (d) 19

31. How many numbers up to 200 are divisible 2 and 3 both?

(a) 35 (b) 33
(c) 29 (d) 27

32. How many numbers between 100 and 300 are divisible by 11?

(a) 11 (b) 10
(c) 12 (d) 18

33. How many numbers between 150 and 500 are divisible by 2, 3 and 7 together?

(a) 9 (b) 8
(c) 10 (d) 11

34. The number of five figures to be added to a number of four fives to obtain the least number of six figures exactly divisible by 357 is:

(a) 94762 (b) 94802
(c) 94485 (d) None of these

35. The nearest figure to 58701 which is divisible by 567 is:

(a) 58968 (b) 58434
(c) 58401 (d) None of these

36. The digits indicated by * in 3422213 ** so that this number is divisible by 99 are:

(a) 1, 9 (b) 3, 7
(c) 4, 6 (d) 5, 5

37. The least value to be given to * so that the number 5 * 3457 is divisible by 11 is:
(a) 2 (b) 3
(c) 0 (d) 4

38. The nearest whole number to one million which is divisible by 537 is:
(a) 1000106 (b) 999894
(c) 1000437 (d) 999563

39. The smallest number between 400 and 500 which is divisible by 9 is:
(a) 414 (b) 405
(c) 423 (d) None of these

40. Which one of the following is the greatest number of five digits divisible by 231?
(a) 99792 (b) 99892
(c) 99692 (d) 99972

41. Find the number nearest to 16386 which is exactly divisible by 425.
(a) 16575 (b) 16375
(c) 16050 (d) 16450

42. Find the least number which must be subtracted from 9269 so that resulting number is exactly divisible by 73?
(a) 17 (b) 57
(c) 71 (d) 63

43. Find the least number which must be added to 15463 so that the resulting number is exactly divisible by 107?
(a) 52 (b) 71
(c) 55 (d) 19

44. What is the number just more than 5000 which is exactly divisible by 73?
(a) 5001 (b) 5009
(c) 5037 (d) 5027

45. The sum of two numbers is 100 and their difference is 37. The difference of their squares is:
(a) 37 (b) 100
(c) 63 (d) 3700

46. The number of times 79 be subtracted from 50000, so that the remainder be 43759; is:
(a) 69 (b) 79
(c) 59 (d) None of these

47. The ratio between two numbers is 3:4 and their sum is 420. The greater of the two numbers is:
(a) 175 (b) 200
(c) 240 (d) 315

48. The difference between the squares of two consecutive numbers is 35. The numbers are:
(a) 14, 15 (b) 15, 16
(c) 17, 18 (d) 18, 19

49. Three-fourths of one-fifth of a number is 60. The number is:
(a) 300 (b) 400
(c) 450 (d) 1200

50. The sum of squares of two numbers is 80 and the square of their difference is 36. The product of the two numbers is:
(a) 22 (b) 44
(c) 58 (d) 116

51. A number when divided by 357 gives a remainder 37. By dividing the same number by 17, the remainder would be:
(a) 3 (b) 4
(c) 2 (d) None of these

52. The product of two numbers is 120. The sum of their squares is 289. The sum of the two numbers is:
(a) 20 (b) 23
(c) 169 (d) None of these

53. Three numbers are in the ratio 4:5:6 and their average is 25. The largest number is:
(a) 42 (b) 36
(c) 30 (d) 32

54. A number exceeds 20% of itself by 40. The number is:
(a) 50 (b) 60
(c) 80 (d) 320

55. If 16% of 40% of a number is 8, the number is:
(a) 200 (b) 225
(c) 125 (d) 320

56. 4767 exactly divides *** 341, the missing digits are:
(a) 468 (b) 586
(c) 363 (d) None of these

57. A number when divided by a certain divisor left remainder 241, when twice the number was divided by the same divisor, the remainder was 112. Find the divisor.
(a) 370 (b) 365
(c) 380 (d) 456

58. Two numbers when divided by a certain divisor give remainders 43 and 37 respectively, when their sum is divided by the same divisor, the remainder is 13. Find the divisor.
(a) 71 (b) 67
(c) 57 (d) 77

59. Two numbers are such that the ratio between them is 3:5; but if each is increased by 10, the ratio between them becomes 5:7. The numbers are:
(a) 3, 5 (b) 7, 9
(c) 13, 22 (d) 15, 25

60. Divide 50 into two parts so that the sum of their reciprocals is 1/12.
(a) 20, 30 (b) 24, 26
(c) 28, 22 (d) 36, 14

61. The sum of seven numbers is 235. The average of the first three is 23 and that of last three is 42. The fourth number is:
(a) 40 (b) 126
(c) 69 (d) 195

62. The sum of squares of two numbers is 68 and the squares of their difference is 36. The product of the two numbers is:
(a) 16 (b) 32
(c) 58 (d) 104

63. What is the least value of K so that the number 6735K1 is divisible by 9?
(a) 5 (b) 7
(c) 4 (d) 3

64. For what value of K, the number 7236K2 is divisible by 8?
(a) 7 (b) 5
(c) 4 (d) 9

65. Find the least values of x and y so that the number 5x423y is divisible by 88.
(a) 8, 2 (b) 7, 3
(c) 9, 4 (d) 6, 5

66. 24 is divided into two parts such that 7 times the first part added to 5 times the second part makes 146. The first part is:
(a) 13 (b) 15
(c) 17 (d) 19

67. Sum of three numbers is 132. First number is twice the second and third number is one-third of the first. Find the second number.
(a) 18 (b) 36
(c) 20 (d) 16

68. What least number must be added to 7231 so that the resulting number is exactly divisible by 5 and 9 together?
(a) 20 (b) 18
(c) 14 (d) 16

69. Find a number nearest to 9231 which is exactly divisible by 3 as well as by 11.
(a) 9240 (b) 9340
(c) 9540 (d) 9440

70. Find a nearest number to 12199 which is exactly divisible by the product of the first four prime numbers.
(a) 12181 (b) 12179
(c) 11281 (d) 11279

71. The sum of squares of two numbers is 90 and the squares of their difference is 46. The product of the two numbers is:
(a) 22 (b) 24
(c) 26 (d) 28

72. If 40% of a number is 360, what will be 15% of that number?
(a) $20\frac{1}{4}$ (b) $20\frac{1}{2}$
(c) $22\frac{1}{4}$ (d) $22\frac{1}{2}$

73. The sum of the digits of a two-digit number is 8. If the digits are reversed the number is increased by 54. Find the number.
(a) 17 (b) 19
(c) 21 (d) 23

EXERCISE-2
(BASED ON MEMORY)

1. N is the largest two digit number, which when divided by 3, 4 and 6 leaves the remainder 1, 2 and 4 respectively, What is the remainder when N is divided by 5?
(a) 4 (b) 2
(c) 0 (d) 1
[SSC CGL Tier-II CBE, 2018]

2. A and B are positive integers. If $A + B + AB = 65$ then What is the difference between A and B (A, $B \leq 15$)?
(a) 3 (b) 4
(c) 5 (d) 6
[SSC CGL Tier-II CBE, 2018]

3. Which one is the largest among the fractions $\left(\frac{5}{113}\right),\left(\frac{7}{120}\right),\left(\frac{13}{145}\right)$ and $\left(\frac{17}{160}\right)$?

(a) $\frac{5}{113}$ (b) $\frac{7}{120}$

(c) $\frac{13}{145}$ (d) $\frac{17}{160}$

[SSC CGL Tier-I CBE, 2018]

4. On dividing a number by 38, we get 90 as quotient and 19 as remainder. What is the number?

(a) 3401 (b) 3382

(c) 3458 (d) 3439

[SSC CHSL (10 + 2) Tier-I CBE, 2017]

5. (90 + 92 + 93 + ...+ 110) is equal to

(a) 4020 (b) 2010

(c) 6030 (d) 8040

[SSC CGL Tier-I CBE, 2017]

6. If the number 583-437 is completely divisible by 9, then the smallest whole number in the place of the blank digit will be

(a) 4 (b) 5

(c) 3 (d) 6

[SSC CHSL (10 + 2) Tier-I CBE, 2017]

7. The least number of five digits exactly divisible by 88 is:

(a) 10032 (b) 10132

(c) 10088 (d) 10023

[SSC Matric Level MTS, 2017]

8. If 34N is divisible by 11, then what is the value of N?

(a) 1 (b) 3

(c) 4 (d) 9

[SSC CAPFs ASI and Delhi Police SI, 2017]

9. How many positive factors of 36 are there?

(a) 4 (b) 6

(c) 9 (d) 12

[SSC CAPFs ASI and Delhi Police SI, 2017]

10. A boy added all natural numbers from 1 to 12, however he added one number twice due to which the sum became 80. What is the number which he added twice?

(a) 3 (b) 2

(c) 7 (d) 8

[SSC CAPFs ASI and Delhi Police SI, 2017]

11. How many numbers are there between 1 and 200 which are divisible by 3 but not by 7?

(a) 38 (b) 45

(c) 57 (d) 66

[SSC CHSL (10 + 2) Tier-I CBE, 2017]

12. What least number be subtracted from 3401, so that the sum is completely divisible by 11?

(a) 3 (b) 1

(c) 2 (d) 0

[SSC CGL Tier-I CBE, 2017]

13. What is the remainder when 2468 is divided by 37?

(a) 26 (b) 36

(c) 18 (d) 14

[SSC CHSL (10 + 2) Tier-I CBE, 2017]

14. If 432 PI is completely divisible by 9, then what is the value of *P*?

(a) 7 (b) 8

(c) 7 (d) 2

[SSC Multi-Tasking Start, 2017]

15. For what value of M. 34 M is divisible by 12?

(a) 2 (b) 0

(c) 8 (d) 6

[SSC Delhi police Constable, 2017]

16. How many positive factors of 160 are there?

(a) 6 (b) 8

(c) 5 (d) 12

[SSC Multi-Tasking Staff, 2017]

17. Which fraction among $\frac{2}{3},\frac{4}{5}$ and $\frac{7}{11}$ is the largest?

(a) $\frac{2}{3}$ (b) $\frac{4}{5}$

(c) $\frac{7}{11}$ (d) All are equal

[SSC Multi-Tasking Staff, 2017]

18. Which fraction among $\frac{3}{7},\frac{5}{11}$ and $\frac{6}{13}$ is largest ?

(a) $\frac{3}{7}$ (b) $\frac{5}{11}$

(c) $\frac{6}{13}$ (d) All are equal

[SSC Multi-Tasking Staff, 2017]

19. What is the largest two digits prime number?

(a) 91 (b) 93
(c) 97 (d) 99

[SSC Delhi Police constable, 2017]

20. A mason can build a wall in 70 hours. After 7 hours he takes a break. What fraction of the wall is yet to be build?

(a) 0.9 (b) 0.8
(c) 0.5 (d) 0.75

[SSC CHSL (10 + 2) Tier-I (CBE), 2017]

21. Two baskets together have 640 oranges. If $\left(\frac{1}{5}\right)$th of the oranges in the first basket be taken to the second baskets became then, no. of oranges in both baskets equal. The number of oranges in the first basket is

(a) 800 (b) 600
(c) 400 (5) 300

[SSC CGL Tier-II (CBE), 2017]

22. What is the value of $\left(\frac{3}{5}+\frac{7}{9}\right)$?

(a) $\frac{62}{45}$ (b) $\frac{31}{28}$
(c) $\frac{5}{7}$ (d) $\frac{1}{7}$

[SSC CHSL (10 + 2) Tier-I CBE, 2017]

23. The sum of a non-zero number and twice its reciprocal is $\frac{33}{4}$ Find the number.

(a) 9 (b) 10
(c) 11 (d) 8

[SSC CGL Tier-I CBE, 2017]

24. The reciprocal of the sum of the reciprocals of $\frac{6}{5}$ and $\frac{3}{7}$

(a) $\frac{57}{18}$ (b) $\frac{35}{57}$
(c) $\frac{18}{57}$ (d) $\frac{57}{35}$

[SSC CGL Tier-I CBE, 2017]

25. In a college, $\frac{1}{5}$th of the girls and $\frac{1}{8}$th of the boys took part in a social camp. The part of students in the college who took part in the camp is:

(a) $\frac{13}{80}$ (b) $\frac{2}{13}$
(c) $\frac{13}{40}$ (d) $\frac{4}{15}$

[SSC Matric Level MTS, 2017]

26. A fraction is greater than twice its reciprocal by $\frac{7}{15}$. What is the fraction?

(a) $\frac{3}{5}$ (b) $\frac{5}{3}$
(c) $\frac{3}{4}$ (d) $\frac{4}{3}$

[SSC CGL Tier-I CBE, 2017]

27. A fraction is greater than its reciprocal by $\frac{9}{20}$. What is the fraction?

(a) $\frac{5}{4}$ (b) $\frac{4}{5}$
(c) $\frac{3}{4}$ (d) $\frac{4}{3}$

[SSC CGL Tier-I CBE, 2017]

28. A number is greater than 58 times its reciprocal by $\frac{3}{4}$. What is the number?

(a) –8 (b) 12
(c) –12 (d) 8

[SSC CGL Tier-I CBE, 2017]

29. Natu and Buchku each have certain number of oranges. Natu says to Buchku. "If you give me 10 of your oranges. I will have twice the number of oranges left with you". Buchku replies, "If you give me 10 of your oranges, I will have the same number of oranges as left with you". What is the number of oranges with Natu and Buchku respectively?

(a) 50, 20 (b) 70, 50
(c) 20, 50 (d) 50, 70

[SSC CGL Tier-II (CBE), 2017]

30. The ten's digit of a two-digit number is greater than the unit's digit by 7. If we subtract 63 from the number. the new number obtained is a number formed by interchange of the digits. Find the number.

(a) 81 (b) 18
(c) 63 (d) 26

[SSC CGL Tier-I CBE, 2017]

31. The numerator of a fraction is multiple of two numbers. One of the numbers is greater than the other by 2. The greater number is smaller than the denominator $7 + c(c > -7)$ is a constant, then the ninimum value of the fraction is

(a) 5 (b) $\frac{1}{5}$

(c) –5 (d) $\frac{-1}{5}$

[SSC CGL Tier-II (CBE), 2016]

32. In a division sum, the divisor 'd' is 10 times the quotient 'q' and 5 times the remainder 'r'. If r = 46 the dividend will be

(a) 5042 (b) 5328
(c) 5336 (d) 4276

[SSC CGL Tier-II, 2016]

33. A number when divided by 44, gives 432 as quotient and 0 as remainder. What will be the remainder when dividing the same number by 31?

(a) 3 (b) 4
(c) 5 (d) 6

[SSC CPO ASI, 2016]

34. A number when divide by 729 gives a remainder if the same number is divided by 27?

(a) 4 (b) 2
(c) 0 (d) 1

[SSC CPO ASI, 2016]

35. What is the smallest 6-digit number that is completely divisible by 108?

(a) 100003 (b) 100004
(c) 100006 (d) 100008

[SSC CPO, 2016]

36. When a number is divided by 56, the remainder will be 29. If the same number is divided by 8 then the remainder will be

(a) 6 (b) 7
(c) 5 (d) 3

[SSC CGL Tier-I (CBE), 2016]

37. A number when divided by the sum of 555 and 445 gives two times their difference as quotient and 30 as the remainder. The number is

(a) 220030 (b) 22030
(c) 1220 (d) 1250

[SSC CGL Tier-II (CBE), 2016]

38. The sum of three numbers is 2 the 1st number is $\frac{1}{2}$ times the 2nd number and the 3rd number is $\frac{1}{4}$ times the 2nd number. The 2nd number is

(a) $\frac{7}{6}$ (b) $\frac{8}{7}$

(c) $\frac{9}{8}$ (d) $\frac{10}{9}$

[SSC CGL Tier-II CBE, 2016]

39. If $\frac{1}{2}$ is added to a number and the sum is multiplied by 3, the result is 21. Then the number is:-

(a) 6.5 (b) 5.5
(c) 4.5 (d) –6.5

[SSC CGL Tier-I (CBE), 2016]

40. The value of x in the following equation is:

$0.3 + 0.6 + 0.7 + 0.8 = x$

(a) 5.3 (b) $2\frac{3}{10}$

(c) $2\frac{2}{5}$ (d) 2.35

[SSC CAPFs (CPO) SI and ASI. Delhi, 2016]

41. The least number which when divided by 6, 9,12, 15, 18 leaves the same remainder 2 in each case is:

(a) 178 (b) 182
(c) 176 (d) 180

[SSC, 2015]

42. If 3/4 of a number is 7 more than 1/6 of the number, then 5/3 of the number is:

(a) 15 (b) 18
(c) 12 (d) 20

[SSC, 2015]

43. What is the arithmetic mean of first 20 odd natural numbers?

(a) 17 (b) 19
(c) 22 (d) 20

[SSC, 2015]

44. What is the Arithmetic mean of the first 'n' natural numbers?

(a) $\frac{n+1}{2}$ (b) $\frac{n^2(n+1)}{2}$

(c) 2(n + 1) (d) $\frac{n(n+1)}{2}$

[SSC, 2015]

45. There is a number consisting of two digits, the digit in the units place is twice that in the tens place and if 2 be subtracted from the sum of the digits, the difference is equal to $1/6^{th}$ of the number. The number is

(a) 23 (b) 24
(c) 25 (d) 26

[SSC, 2015]

46. Let x be the smallest number, which when added to 2000 makes the resulting number divisible by 12, 16, 18 and 21. The sum of the digits of x is

(a) 7 (b) 4
(c) 6 (d) 5

[SSC, 2015]

47. The unit digit in the product $(2467)^{153} \times (341)^{72}$ is

(a) 3 (b) 9
(c) 7 (d) 1

[SSC, 2015]

48. Let x be the least number, which when divided by 5, 6, 7 and 8 leaves a remainder 3 in each case but when divided by 9 leaves no remainder. The sum of digits of x is

(a) 22 (b) 18
(c) 21 (d) 24

[SSC, 2015]

49. A number when divided by 361 gives a remainder 47. If the same number is divided by 19, the remainder obtained is

(a) 8 (b) 9
(c) 1 (d) 3

[SSC, 2015]

50. The greatest number among 3^{50}, 4^{40}, 5^{30} and 6^{20} is

(a) 6^{20} (b) 3^{50}
(c) 4^{40} (d) 5^{30}

[SSC, 2015]

51. A number exceeds its two fifth by 75. The number is

(a) 100 (b) 112
(c) 150 (d) 125

[SSC, 2015]

52. If the sum of two numbers, one of which is $\frac{2}{5}$ times the other is 50, then the numbers are:

(a) $\frac{250}{7}, \frac{100}{7}$ (b) $\frac{115}{7}, \frac{235}{7}$
(c) $\frac{150}{7}, \frac{200}{7}$ (d) $\frac{240}{7}, \frac{110}{7}$

[SSC, 2015]

53. If $x[-2\{-4(-a)\}] + 5\ [-2\{-2(-a)\}] = 4a$, then x =

(a) –2 (b) –3
(c) –4 (d) –5

[SSC, 2014]

54. In a Mathematics examination the numbers scored by 5 candidates are 5 successive odd integers. If their total marks is 185, the highest score is

(a) 39 (b) 43
(c) 41 (d) 37

[SSC, 2014]

55. Arranging the following in descending order, we get $\sqrt[3]{4}, \sqrt{2}, \sqrt[6]{3}, \sqrt[4]{5}$

(a) $\sqrt[3]{4} > \sqrt[4]{5} > \sqrt{2} > \sqrt[6]{3}$
(b) $\sqrt[3]{4} > \sqrt[4]{5} > \sqrt{2} > \sqrt[6]{3}$
(c) $\sqrt{2} > \sqrt[6]{3} > \sqrt[3]{4} > \sqrt[4]{5}$
(d) $\sqrt[6]{3} > \sqrt[4]{5} > \sqrt[3]{4} > \sqrt{2}$

[SSC, 2014]

56. For any integral value of n, $3^{2n} + 9n + 5$ when divided by 3 will leave the remainder

(a) 1 (b) 2
(c) 0 (d) 5

[SSC, 2014]

57. The least prime number is

(a) 3 (b) 2
(c) 0 (d) 1

[SSC, 2014]

58. Find the least number which when divided by 12, 18, 36 and 45 leaves the remainder 8, 14, 32 and 41 respectively.

(a) 178 (b) 186
(c) 176 (d) 180

[SSC, 2014]

59. The value of 204×197 is

(a) 39812 (b) 40218
(c) 40188 (d) 40212

[SSC, 2014]

60. A piece of cloth measured with a metre stick, one cm short, is 100 metres long. Reckoning the metre stick as being right, the actual length of the cloth (in cm) is

(a) 8,000 (b) 6,100
(c) 3,900 (d) 9,900

[SSC, 2014]

61. The value of (1001)3 is

(a) 100300301 (b) 103003001
(c) 1003003001 (d) 100303001

[SSC, 2014]

62. When 335 is added to 5*A*7, the result is 8*B*2. 8*B*2 is divisible by 3. What is the largest possible value of *A*?

(a) 8 (b) 2
(c) 1 (d) 4

[SSC, 2013]

63. Which one of the following numbers is divisible by 25?

(a) 303310 (b) 373355
(c) 303375 (d) 22040

[SSC, 2013]

64. The units digit in $3 \times 38 \times 537 \times 1256$ is:

(a) 4 (b) 2
(c) 6 (d) 8

[SSC, 2013]

65. A number is of two digits. The position of digits is interchanged and the new number is added to the original number. The resultant number will always be divisible by:

(a) 8 (b) 9
(c) 10 (d) 11

[UPPCS, 2012]

66. The last digit, that is, the digit in the unit's place of the number $[(57)^{25} - 1]$ is:

(a) 6 (b) 8
(c) 0 (d) 5

[SSC Assistant Grade III, 2012]

67. A number N is a positive three-digit number. If x is in its hundred's place and y is in its unit's place, then the number $N - 100x - y$ is always divisible by:

(a) 8 (b) 9
(c) 10 (d) 11

[SSC Assistant Grade III, 2012]

68. n is a whole number which when divided by 4 gives the remainder 3. The remainder when $2n$ is divided by 4 is:

(a) 1 (b) 2
(c) 3 (d) 0

[SSC, 2012]

69. If m and n are positive integers and $(m - n)$ is an even number, then $(m^2 - n^2)$ will be always divisible by:

(a) 4 (b) 6
(c) 8 (d) 12

[SSC, 2012]

70. Rachita enters a shop to buy ice-creams, cookies and pastries. She has to buy at least 9 units of each. She buys more cookies than ice-creams and more pastries than cookies. She picks up a total of 32 items. How many cookies does she buy?

(a) Either 12 or 13 (b) Either 11 or 12
(c) Either 10 or 11 (d) Either 9 or 11
(e) Either 9 or 10

[IBPS PO/MT, 2012]

71. The product of three consecutive even number is 4032. The product of the first and the third number is 252. What is five times the second number?

(a) 80 (b) 100
(c) 60 (d) 70
(e) 90

[IBPS PO/MT, 2012]

72. When 2^{23} is divided by 10, the remainder will be:

(a) 2 (b) 3
(c) 4 (d) 8

[SSC (GL), 2011]

73. Find the unit digit in the product $(4387)^{245} \times (621)^{72}$.

(a) 1 (b) 2
(c) 5 (d) 7

[SSC (GL), 2011]

74. If a and b are odd numbers, then which of the following is even?

(a) $a + b + ab$ (b) $a + b - 1$
(c) $a + b + 1$ (d) $a + b + 2ab$

[SSC (GL), 2011]

75. $2^{16} - 1$ is divisible by:

(a) 11 (b) 13
(c) 17 (d) 19

[SSC (GL), 2011]

76. The sum of two numbers is 24 and their product is 143. The sum of their squares is:

(a) 296 (b) 295
(c) 290 (d) 228

[SSC (GL), 2011]

77. The unit digit in the sum $(124)^{372} + (124)^{373}$ is:

(a) 5 (b) 4
(c) 2 (d) 0

[SSC (GL), 2011]

78. If the sum of two numbers be multiplied by each number separately, the products so obtained are 247 and 114. The sum of the numbers is:

(a) 19 (b) 20
(c) 21 (d) 23

[SSC (GL), 2011]

79. Find a number, one-seventh of which exceeds its eleventh part by 100.

(a) 1925 (b) 1825
(c) 1540 (d) 1340

[SSC (GL), 2011]

80. $\sqrt{6+\sqrt{6+\sqrt{6+\cdots=?}}}$

(a) 2.3 (b) 3
(c) 6 (d) 6.3

[SSC (GL), 2011]

81. In a division sum, the divisor is 4 times the quotient and twice the remainder. If a and b are respectively the divisor and the dividend, then:

(a) $\frac{4b-a^2}{a}=3$ (b) $\frac{4b-2a}{a^2}=2$

(c) $(a+1)^2=4b$ (d) $\frac{a(a+2)}{b}=4$

[SSC, 2011]

82. If 738*A*6*A* is divisible by 11, then the value of A is:

(a) 6 (b) 3
(c) 9 (d) 1

[SSC, 2011]

83. The product of two numbers is 1575 and their quotient is $\frac{9}{7}$. Then the sum of the numbers is:

(a) 74 (b) 78
(c) 80 (d) 90

[SSC, 2011]

84. When $(67^{67}+67)$ is divided by 68, the remainder is:

(a) 1 (b) 63
(c) 66 (d) 67

[SSC, 2011]

85. The least positive integer that should be subtracted from 3011 × 3012 so that the difference is a perfect square, is:

(a) 3009 (b) 3010
(c) 3011 (d) 3012

[SSC, 2011]

86. The sum of nine consecutive odd numbers of Set A is 621. What is the sum of a different set of six consecutive even numbers whose lowest number is 15 more than the lowest number of Set A?

(a) 498 (b) 468
(c) 478 (d) 488
(e) None of these

[IOB PO, 2011]

87. The sum of the squares of two consecutive even numbers is 6500. Which is the smaller number?

(a) 54 (b) 52
(c) 48 (d) 56

[Punjab National Bank PO, 2010]

88. The sum of five consecutive even numbers of set-A is 220. What is the sum of a different set of five consecutive numbers whose second lowest number in 37 less than double of the lowest number of set-A?

(a) 223 (b) 225
(c) 235 (d) None of these

[CBI PO, 2010]

89. A number, when divided by 296, gives 75 as the remainder. If the same number is divided by 37 then the remainder will be:

(a) 1 (b) 2
(c) 19 (d) 31

[SSC, 2010]

90. The sum and product of two numbers are 12 and 35 respectively. The sum of their reciprocals will be:

(a) $\frac{1}{3}$ (b) $\frac{1}{5}$

(c) $\frac{12}{35}$ (d) $\frac{35}{12}$

[SSC, 2010]

91. The greatest number among $\sqrt{5}, \sqrt[3]{4}, \sqrt[5]{2}, \sqrt[7]{3}$ is:

(a) $\sqrt[3]{4}$ (b) $\sqrt[7]{3}$

(c) $\sqrt{5}$ (d) $\sqrt[5]{2}$

[SSC, 2010]

92. Two numbers are such that their difference, their sum and their product are in the ratio of 1:7:24. The product of the numbers is:

(a) 24 (b) 36
(c) 48 (d) 60

[SSC, 2010]